

# Souvik Mukherjee

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[LinkedIn](#) | [Github](#) | [Portfolio](#) | [Codeforces](#) | [LeetCode](#)

## Summary

Computer Science engineer specializing in **distributed systems**, **concurrency**, and **systems programming** in **Go** and **C++**. Builds production-grade infrastructure from first principles — TCP servers, compilers, and version-control systems — with strong foundations in **data structures**, **algorithms**, and **software design**. GATE CS **AIR 48** (2026, top 0.02%) and Codeforces **Expert**.

## Education

|                                                         |                     |                      |
|---------------------------------------------------------|---------------------|----------------------|
| <b>B.Tech in Computer Science (2022–2026)</b>           | Calcutta University | <b>CGPA: 9.61/10</b> |
| <b>M.Tech in Computer Science (Incoming, 2026–2028)</b> | IIT Madras          |                      |
| <b>Class 12 (CBSE), 2022</b>                            | Green Point Academy | <b>95.33%</b>        |

## Experience

- Extended Dynamic Solutions (Kolkata, India)** Nov 2025 – May 2026  
*Backend Intern* **Tech Stack:** Go (Golang), AWS (Cognito, SQS, S3), gRPC, REST, Terraform, CI/CD
  - Took ownership of designing a **scalable distributed, microservices architecture (4+ services)** with **high availability** and **concurrency-safe** request handling; dove deep into access control to implement a granular RBAC **software design** controlling fine-grained permissions across **6 roles**.
  - Engineered the integration of **AWS Cognito** for passwordless authentication and **AWS S3** for KYC storage; developed **30+ REST APIs** and **15+ gRPC** methods, implementing explicit exception-mapping handlers to improve error **reliability** and debuggability across services.
  - Architected an **event-driven, asynchronous system** using **AWS SQS** with multi-worker task processing, decoupling critical workflows and improving **fault tolerance**; automated infrastructure via **Terraform (IaC)** and built end-to-end **CI/CD** pipelines with GitHub Actions.

## Projects

- GORedis — Redis-Compatible, Concurrent In-Memory Database** [github.com/Souvik606/Go-redis](https://github.com/Souvik606/Go-redis)  
**Tech Stack:** Go (Golang)
  - Architected a **multi-threaded, concurrent TCP server** from scratch using goroutines and **synchronization primitives** (mutexes, atomics), handling thousands of multiplexed connections and **30+ RESP commands** across **5 data types**.
  - Implemented **atomic transactions (MULTI/EXEC)**, lazy & active key expiration, and a thread-safe Pub/Sub system with dynamic channel subscriptions — engineered to avoid race conditions under concurrent load.
  - Built an async **Append-Only File** persistence layer with background compaction; profiled and benchmarked **performance** against Redis v7.0.15, achieving **37K+ ops/sec** at **0.83ms** avg latency under 50 concurrent clients.
- Goit — Distributed Version Control System** [github.com/Souvik606/goit](https://github.com/Souvik606/goit)  
**Tech Stack:** Go (Golang)
  - Designed a **DAG-based** state machine and a custom **Diff3 algorithm** for 3-way merges, applying core **algorithms and data structures** to traverse history at **5,000 commits/ms**.
  - Built **20+ CLI commands** and a custom distributed HTTP server (**push/pull/clone**) sustaining **150 MB/s** clone throughput; designed a **zlib-compressed** object store with **8,000 files/sec** staging.
- Sardine — A Compiler Frontend & AST-Based Interpreter** [github.com/Souvik606/Mission-Sardine](https://github.com/Souvik606/Mission-Sardine)  
**Tech Stack:** C++23, CMake, Next.js, Tailwind CSS
  - Built a **compiler frontend** from scratch: a custom **DFA lexical scanner** and **Recursive Descent parser** feeding a two-pass tree-walk interpreter supporting **40+ language constructs** across procedural and OOP paradigms.
  - Profiled **performance** against Python 3.10 — lower peak memory (5–8 MB vs. 9–10 MB) and near-parity on floating-point math; **debugged** AST-traversal bottlenecks causing **13.97x** slower recursive execution (Fibonacci), informing optimization priorities.
  - Designed the **Sardine Web Console**, a **visualization** tool that parses serialized pipeline telemetry into JSON to demonstrate AST generation and execution flow in real time.

## Achievements

- Cracked **GATE CS** with **AIR 48** in 2026 (top 0.02% of 211,000 aspirants) and **AIR 199** in 2025 out of 170,000 aspirants.
- Attained the title of **Expert** (peak rating **1729**) on **Codeforces** under handle **v1kCoder**; **Knight** badge on **LeetCode** (peak rating **1885**).
- Secured global rank **412** in Codeforces Round 1043 (Div.3) and **756** in Round 1049 (Div.2).

## Co-Curriculars & Extra-Curriculars

- Co-organized **Code Combat**, a multi-tier competitive programming tournament at **Reflexons 2026** (CU Technology Campus), designing problem distribution metrics and managing contest logistics.
- Secured **1st place** in the Inter-College Treasure Hunt at the same departmental fest.

## Skills

|                           |                                                                         |
|---------------------------|-------------------------------------------------------------------------|
| <b>Languages:</b>         | Go, C++, Python, JavaScript; Familiar with TypeScript, C, Java          |
| <b>Frameworks:</b>        | Node.js, React.js, Next.js, Tailwind CSS, ShadCN, OpenCV                |
| <b>Tools:</b>             | Git, GitHub, Docker, VS Code, Postman, Prisma, Terraform                |
| <b>Databases:</b>         | MySQL, PostgreSQL, MongoDB                                              |
| <b>Cloud &amp; Infra:</b> | AWS (Cognito, SQS, S3), gRPC, OpenTelemetry, CI/CD, Microservices, REST |